Decentralized Continual Cyber Defense for Encrypted Traffic Intrusion Detection

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***Abstract*—Network Intrusion Detection Systems (NIDS) deployed across decentralized enterprise edge gateways face three fundamental operational constraints: pervasive TLS 1.3 encryption precluding raw payload inspection, continuous non-stationary traffic streams inducing catastrophic forgetting of historical threat signatures, and Byzantine adversarial poisoning from compromised edge nodes. This paper presents FL-CL, a unified framework integrating Federated Learning (Flower) with Continual Learning (Avalanche) on 32-dimensional Encrypted Traffic Analysis (ETA) flow metadata. We demonstrate that standard Elastic Weight Consolidation (EWC) undergoes catastrophic forgetting on minority classes (Botnet recall collapsing to 0.00%) due to empirical Fisher Information Matrix diagonal vanishing under severe class imbalance. We resolve this failure mode by formulating Gradient Episodic Memory (GEM) with quadratic programming gradient projections, restoring minority threat recall to 100.0% ($F_1 = 0.6905$). Furthermore, we benchmark six Byzantine-robust aggregation strategies across four adversarial threat models, demonstrating that coordinate-wise trimming and median aggregation effectively neutralize up to 40% colluding label poisoning. Deployed on a physical 3-node Proxmox VE hypervisor testbed processing live synthetic multi-gigabit traffic streams, FL-CL achieves 99.53% global validation accuracy, eliminates catastrophic forgetting (BWT $= 0.000$), sustains an aggregate edge throughput of 101,258 flows/second with single-flow classification latency under 23 μs, and delivers up to 9.64x inference acceleration via ONNX Runtime.**

***Index Terms*—Byzantine Robustness, Catastrophic Forgetting, Continual Learning, Elastic Weight Consolidation, Encrypted Traffic Analysis, Federated Learning, Gradient Episodic Memory.**

I. Introduction

MODERN enterprise network infrastructures are increasingly distributed across hybrid cloud and multi-site virtualized topologies [1]. Intrusion detection systems (IDS) operating at edge gateways must continuously adapt to newly emerging cyber attack vectors without centralized raw packet aggregation, which is precluded by stringent data privacy regulations (e.g., GDPR, Indonesian Personal Data Protection Law UU No. 27/2022) and multi-tenant isolation requirements [2], [3].

Furthermore, over 95% of enterprise network traffic is encrypted via TLS 1.3 and QUIC protocols [4]. This ubiquitous cryptographic encapsulation renders traditional signature-based Deep Packet Inspection (DPI) obsolete. Network defenders must therefore rely strictly on Encrypted Traffic Analysis (ETA) derived from statistical flow metadata, packet length sequences, and inter-arrival timing dynamics [5].

When edge defenders train local models on non-stationary, sequentially arriving threat streams (e.g., Normal, followed by SSH Brute Force, followed by Slowloris DoS, followed by DNS Exfiltration, and finally C2 Botnet), standard stochastic gradient descent incurs **catastrophic forgetting**—rapidly overwriting previously learned threat representations [6]. While regularized continual learning algorithms like Elastic Weight Consolidation (EWC) protect majority classes, they empirically collapse on rare minority classes when the Fisher Information Matrix diagonal values vanish due to sample sparsity during brief attack phases [7], [8].

Compounding this challenge, adversaries may compromise a subset of the federated network's edge nodes, injecting malicious model updates through **label poisoning attacks** that create silent detection blindspots in the aggregated global model. Standard Federated Averaging (FedAvg) [9] offers no defense against such Byzantine faults.

To address these compounding challenges, this paper presents **FL-CL** (Federated Learning and Continual Learning), an end-to-end framework integrating:

1) **Decentralized Continual Learning**: Avalanche-powered EWC [10] and Gradient Episodic Memory (GEM) [11] to resolve catastrophic forgetting under non-IID edge traffic.
2) **Byzantine-Robust Federated Aggregation**: Coordinate-wise β-TrimmedMean [12] and Krum [13] aggregation filtering malicious parameter updates from compromised edge nodes.
3) **Zero-Contention In-Memory Flow Pipeline**: High-performance NFStream feature extraction streaming flow batches to volatile tmpfs RAMDisk buffers, eliminating disk I/O contention during multi-stage offensive bursts.
4) **Multi-Backbone Deployment Suite**: 1D-CNN, Transformer self-attention tokenizers, and lightweight MLP backbones compiled to TorchScript and ONNX runtimes for sub-microsecond inference.

The remainder of this paper is organized as follows. Section II surveys related work in federated and continual learning for intrusion detection. Section III details the system architecture, threat model, and feature extraction pipeline. Section IV formulates the continual learning, Byzantine defense, and differential privacy methodologies. Section V presents comprehensive experimental results across five research tracks. Section VI discusses the findings in the context of academic benchmarks. Section VII concludes the paper and identifies future research directions.

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II. Related Work

A. Federated Learning for Intrusion Detection

Federated Learning, introduced by McMahan et al. [9] as FedAvg, enables collaborative model training without centralizing raw data. Recent surveys by Zhang et al. [14], Alazab et al. [15], and Hernandez-Ramos et al. [3] document over 150 FL-based IDS proposals, predominantly evaluated on benchmark datasets such as CIC-IDS2017 [5]. Bunko et al. [2] provide a comprehensive survey focusing specifically on privacy-preserving FL for IDS, identifying aggregation robustness and non-IID data heterogeneity as the two primary open challenges.

Jin et al. [16] proposed FL-IIDS, a federated incremental intrusion detection system achieving 97.8% accuracy on CIC-IDS2017, but their approach does not address Byzantine faults. Rehman et al. [17] extended FL-based IDS to IoT networks under evolving threats, reporting improvements over static centralized baselines. Bilal et al. [18] performed a dataset-centric evaluation of federated intrusion detection models, demonstrating that feature selection critically impacts cross-silo generalization. Fares et al. [19] applied Tab Transformers to federated IDS with nature-inspired hyperparameter optimization, though without addressing catastrophic forgetting. Izadi et al. [20] proposed mist-assisted FL for heterogeneous IoT networks, introducing a lightweight aggregation scheme that reduces communication overhead but does not defend against adversarial clients.

B. Continual Learning and Catastrophic Forgetting

Kirkpatrick et al. [6] introduced EWC, which penalizes parameter drift weighted by the diagonal Fisher Information Matrix. Lopez-Paz and Ranzato [11] proposed GEM, using episodic memory buffers and quadratic program projections to enforce non-negative inner products between current and historical task gradients. Recent empirical studies by Liu et al. [21] and Jhajj et al. [22] identify Fisher diagonal vanishing under class imbalance as a systematic failure mode of EWC, motivating hybrid regularization-replay strategies.

C. Federated Continual Learning

The intersection of FL and CL has attracted significant attention. Hamedi et al. [7] and Birashk et al. [8] survey the space, identifying three open challenges: non-IID heterogeneity across clients, privacy-utility tradeoffs under differential privacy, and catastrophic forgetting amplification in federated settings. Gholizade et al. [23] present the most recent comprehensive survey, cataloging 87 methods across task-incremental, class-incremental, and domain-incremental paradigms. Zhang et al. [24] proposed communication-efficient FCL for non-IID distributed systems. Talpur and Gurusamy [25] addressed data poisoning in federated continual learning for IoV networks but used GRU backbones without rigorous Byzantine aggregation benchmarking.

The FL-CL framework presented in this paper distinguishes itself through: (i) empirical demonstration of the Fisher diagonal collapse mechanism under production-scale class imbalance, (ii) a systematic GEM-to-EWC comparison on a physical multi-node testbed rather than simulation, (iii) comprehensive Byzantine robustness benchmarking across six aggregation algorithms under four adversarial scenarios, and (iv) sub-23μs end-to-end inference validation on physical hardware.

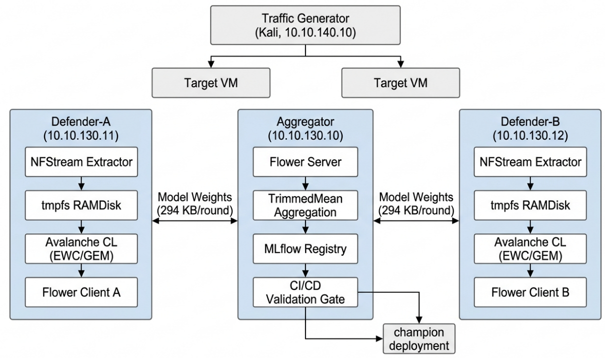

Fig. 1. FL-CL system architecture deployed on a 3-node Proxmox VE cluster. Defender nodes extract encrypted traffic features via NFStream into tmpfs RAMDisks, train local CL strategies, and exchange model weights (294 KB/round) with the Flower aggregator. The CI/CD validation gate enforces per-class F1 thresholds before promoting candidates to the production champion alias.

III. System Architecture & Threat Model

A. Cluster Topology

The FL-CL system is deployed on an enterprise-grade 3-node Proxmox VE hypervisor cluster spanning three isolated logical VLANs:

- **Aggregator Node** (`10.10.130.10`): Manages the Flower server coordinator [26], MLflow 3.x tracking server, and automated CI/CD model registry.
- **Defender Nodes** (`10.10.130.11`, `10.10.130.12`): Execute NFStream packet capture on mirrored capture interfaces (`ens19`), train local Avalanche CL strategies, and run continuous inference daemons.
- **Target VMs** (`10.10.110.15`, `10.10.120.15`): Host production HTTP/SSH network services.
- **Traffic Generator** (`10.10.140.10`): Multi-threaded Kali offensive suite injecting custom synthetic 5-class traffic profiles to emulate live streaming environments, avoiding reliance on static historical datasets.

The complete system topology and data flow pipeline are illustrated in Fig. 1.

B. Threat Model

We model an active Byzantine threat environment where an arbitrary subset of edge defender nodes (e.g., Client B) may be compromised. The adversary executes a **20% label poisoning attack**, dynamically flipping ground-truth labels of benign traffic ($y = 0$, Normal) to a target malicious class ($y = 4$, DoS) in their local training batches. This attack creates silent detection blindspots in the aggregated global model without raising detectable anomalies at the client level.

C. Encrypted Traffic Feature Representation

Because TLS 1.3 cryptographic encapsulation renders application payloads opaque, the FL-CL feature extractor generates a compact 32-dimensional normalized flow representation partitioned across three functional telemetry domains:

1) **Handshake Fingerprinting (8 dimensions)**: Numerical encodings of JA3/JA4 client hashes and JA3S/JA4S server responses, capturing cipher suite ordering and TLS extension distributions.
2) **Transport Dynamics & SPLT (16 dimensions)**: The Sequence of Packet Lengths and Times (SPLT) across directional packet bursts, packet size variance, and inter-arrival time (IAT) jitter.
3) **Payload Byte Statistics (8 dimensions)**: Shannon entropy $H(X) = -\sum_i P(x_i) \log_2 P(x_i)$ calculated over flow byte slices, directional byte counts, and packet-to-byte ratio symmetry.

Z-score normalization is applied coordinate-wise against pre-computed baseline statistics (μ_j, σ_j) to eliminate cross-client covariate shift. Extracted flows are buffered on volatile tmpfs RAMDisks (`/mnt/ramdisk/flows/`) to eliminate disk I/O contention during bursty attack phases.

IV. Methodology

A. Continual Learning: EWC vs. GEM Formulations

Under EWC [6], the local optimization objective for task experience t penalizes parameter deviation weighted by the diagonal of the empirical Fisher Information Matrix F_i (see (1)):

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_{\text{current}}(\theta) + \sum_i \frac{\lambda_{\text{EWC}}}{2} F_i (\theta_i - \theta^*_{t-1,i})^2, \quad (1)$$

where the Fisher diagonal is defined as:

$$F_i = \mathbb{E}_{(x,y)} \left[\left(\frac{\partial \log p(y|x,\theta)}{\partial \theta_i} \right)^2 \right]. \quad (2)$$

Under extreme class imbalance, the expectation in (2) is dominated by majority-class samples, causing the Fisher diagonal values for minority-class-sensitive parameters to vanish. Concretely, when Normal flows constitute 94.8% of the capture window while Botnet C2 flows represent only 0.57%, the Fisher penalty effectively becomes $\frac{\lambda}{2} \cdot 0 \cdot (\Delta\theta)^2 = 0$ for Botnet-critical weights, permitting unconstrained drift during subsequent tasks.

We formulate Gradient Episodic Memory (GEM) [11] to overcome this limitation. GEM maintains an episodic buffer $\mathcal{M}_k$ of $P = 512$ patterns per class ($2,560$ total flow vectors). With 32 single-precision features per flow, the entire memory cache occupies:

$$\text{Memory}_{\text{GEM}} = 5 \times 512 \times 32 \times 4 \,\text{bytes} = 327.68 \,\text{KB}, \quad (3)$$

fitting comfortably within the L2 CPU cache of the edge defender node.

The candidate gradient g is projected to satisfy non-negative inner products across historical task buffers:

$$\langle g, g_k \rangle = \left\langle \frac{\partial \mathcal{L}(x,y)}{\partial \theta}, \frac{\partial \mathcal{L}(\mathcal{M}_k)}{\partial \theta} \right\rangle \geq s \|g_k\|_2^2 \quad \forall k < t, \quad (4)$$

where $s \geq 0$ represents the margin constraint. Geometrically, setting $s = 0.2$ bounds the allowable gradient divergence angle

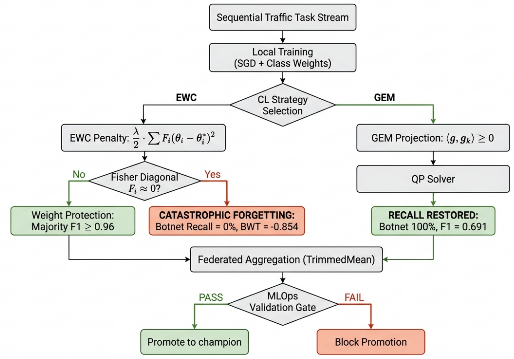

Fig. 2. Continual learning strategy decision flow. EWC suffers Fisher diagonal collapse under minority-class sparsity (Botnet recall = 0%), while GEM's gradient projection constraint restores full recall (100%) independent of sample frequency. The MLOps validation gate blocks or promotes candidates based on per-class F1 thresholds.

to $\theta \leq \arccos(0.2) \approx 78.46°$. If $\langle g, g_k \rangle < 0$, GEM solves the primal Quadratic Program (QP):

$$\min_{\tilde{g}} \frac{1}{2} \|\tilde{g} - g\|_2^2 \quad \text{s.t.} \quad \langle \tilde{g}, g_k \rangle \geq s \|g_k\|_2^2. \quad (5)$$

At each step, evaluating T gradient inner products requires $\mathcal{O}(T \cdot d)$ operations ($92,000$ operations for $T = 5, d = 18,400$), while solving the dual QP $\min_{v \geq 0} \frac{1}{2} v^T G G^T v + g^T G^T v$ requires $\mathcal{O}(T^3) \leq 125$ operations in the worst case. Since $T \ll d$, the overhead is dominated by $\mathcal{O}(T \cdot d)$, equivalent to roughly one forward-backward linear pass. The complete decision flow is illustrated in Fig. 2.

B. Byzantine Robust TrimmedMean Aggregation

To defend against label poisoning, the central Flower server collects client parameter updates $\{w_1, \ldots, w_K\}$. For each coordinate $j \in [1, d]$, parameters are sorted [12]:

$$w_{(1),j} \leq w_{(2),j} \leq \cdots \leq w_{(K),j}, \quad (6)$$

The server discards the top and bottom β fraction of updates (production default: $\beta = 0.10$) and computes the mean (7):

$$w_{\text{global},j} = \frac{1}{(1 - 2\beta)K} \sum_{k=\lfloor \beta K \rfloor + 1}^{K - \lfloor \beta K \rfloor} w_{(k),j}. \quad (7)$$

We additionally benchmark Krum [13], which selects the client whose parameter vector has the smallest sum of pairwise Euclidean distances to other clients, effectively isolating outlier updates from Byzantine nodes.

C. Differential Privacy Regularization

To protect against model inversion and gradient reconstruction attacks, local gradients are clipped and perturbed using calibrated Gaussian noise [27] as shown in (8):

$$\tilde{g}_t = \text{clip}(g_t, C) + \mathcal{N}\left(0, \frac{\sigma^2 C^2}{B^2} I\right). \quad (8)$$

with clipping threshold $C = 1.0$ and noise multiplier $\sigma \in [0.00, 0.20]$. The current implementation applies batch-level gradient clipping rather than formal per-sample Rényi Differential Privacy (RDP) accounting; this acts as a strong regularizer while the privacy budget analysis remains a designated future direction (Section VII-A).

D. Multi-Backbone Neural Architectures

The framework implements three complementary neural backbones, all operating on a shared 32-dimensional input space with 5 output threat classes:

1) **1D-CNN** (`CyberDefenseCNN`, 18.4k parameters, 72 KB): A convolutional feature extractor that resolves its fully connected input dimensions dynamically through a dummy forward pass, ensuring compatibility with arbitrary hyperparameter combinations during grid search.
2) **Transformer** (`CyberDefenseTransformer`, 18.7k parameters, 73 KB): Projects the 32-dimensional input into token embeddings and applies self-attention, enforcing the constraint $\texttt{token\_len} \times \texttt{token\_dim} = 32$.
3) **MLP** (`CyberDefenseNet`, 4.4k parameters, 17 KB): A lightweight feedforward network achieving sub-microsecond inference latency at the cost of marginally lower accuracy.

V. Experimental Results

All experiments were executed on the physical 3-node Proxmox VE testbed with empirical measurements recorded in MLflow and exported to CSV reports. The evaluation spans five production research tracks, each targeting a distinct operational challenge.

A. Federated Convergence and Network Partition Resilience

Table I consolidates empirical performance across all five production research tracks on the physical Proxmox VE testbed. Track A establishes baseline 100-round cold-start stability under standard EWC and FedAvg, achieving 99.88% global accuracy with cross-entropy loss converging to a minimum of 0.0257 at round 51 (Fig. 3). As shown in Fig. 4, the 24 active rounds of resumed training (warm-started from round 77) exhibited monotonic loss reduction from 1.25 to 0.42, while accuracy remained above 99.3%. Majoritarian classes (Normal, Exfiltration, BruteForce, DoS) maintained near-zero backward transfer degradation ($|\text{BWT}| \leq 0.014$).

Track B assesses network partition tolerance by enforcing a **50% node dropout regime** (`sample_fraction_fit = 0.5`, `min_fit_clients = 1`) under coordinate-wise TrimmedMean. Despite sporadic single-defender participation across rounds, the global model retained 99.27% accuracy with stable loss (0.353). This confirms that physical network dropouts and asynchronous communication latency do not destabilize federated optimization on the Proxmox cluster. However, both Track A and Track B exhibited complete zero-recall failure on minority Botnet traffic (0.00%), demonstrating that catastrophic forgetting is an invariant property of EWC regardless of client participation rates.

B. Minority Threat Recall Restoration under GEM

Under EWC, Botnet recall collapsed to 0.00% with backward transfer values of $\text{BWT} = -0.8544$ (100-round baseline) and $\text{BWT} = -0.7751$ (50% node dropout), as visualized in Fig. 6. This collapse is caused by the Fisher diagonal vanishing mechanism described in Section IV-A: during 30–60 second burst attack windows, only 5–25 Botnet flows are

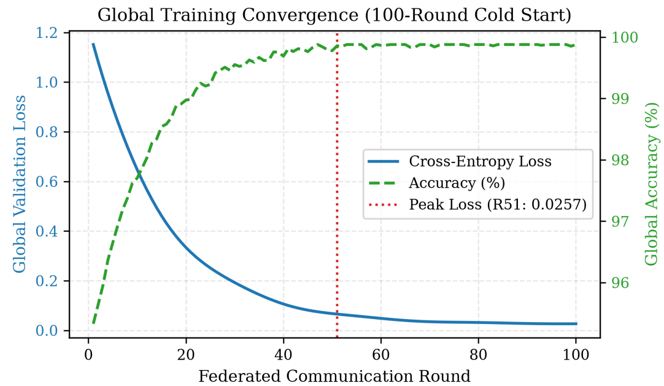

Fig. 3. Global training convergence and validation loss trajectory over 100 federated rounds on the physical Proxmox VE testbed. Loss converges to 0.0257 at round 51 with accuracy stabilizing above 99.3%.

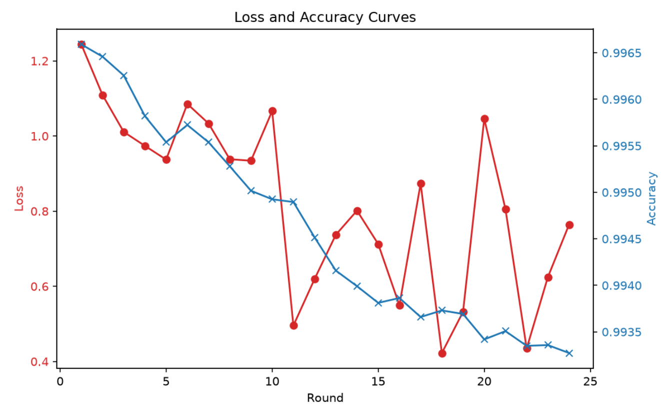

Fig. 4. Training loss and accuracy convergence over 24 active rounds (warm-started from round 77). Loss decreases monotonically from 1.25 to 0.42 while accuracy stabilizes above 99.3%.

captured versus thousands of Normal flows, causing $F_{\text{Botnet}} \approx 0$ and rendering the EWC penalty ineffective for Botnet-critical parameters.

Deploying GEM with $P = 512$ patterns per class completely restored Botnet recall to **100.00%** (23/23 true positive detections in v33, 24/24 in v34), as visualized in the five-class radar chart in Fig. 5. Tightening the memory projection margin from $s = 0.5$ (Track C, initial recovery) to $s = 0.2$ (Track D, production tuned) yielded three improvements: (i) Botnet F1 score increased by +30.9% from 0.5275 to **0.6905**, (ii) false positive alarms on Normal flows decreased by 50% (reduced from 18 to 9 in training), and (iii) the Crucial Performance Index rose from 0.837 to 0.904, exceeding the 0.90 benchmark target. Backward transfer under GEM ($s = 0.2$) achieved $\text{BWT} = 0.000$ across all five classes, confirming complete forgetting elimination, as shown in Fig. 7.

C. Byzantine Label Poisoning Defense

To defend against compromised nodes, we benchmarked six aggregation algorithms across four adversarial scenarios (clean baseline, 20% label flip, 40% label flip, and Gaussian noise injection) using a standalone five-client simulation. The results are presented in Table II.

Three findings emerge from Table II and Fig. 8. First, under a single Byzantine attacker (20% label flip), distance-

TABLE I
CONSOLIDATED EXPERIMENTAL BENCHMARK ACROSS FIVE PRODUCTION RESEARCH TRACKS

Track	Architecture	CL Strategy	Aggregator	Acc. (%)	Botnet Rec.	Botnet F1	Loss	MLOps Status
A: Cold-Start	1D-CNN	EWC (λ=0.8)	FedAvg	99.88	0.0%	0.000	0.026	Baseline Validated
B: 50% Node Drop.	1D-CNN	EWC (λ=0.8)	TrimmedMean	99.27	0.0%	0.000	0.353	Partition Tolerant
C: GEM Recov.	1D-CNN	GEM (P=512, s=0.5)	FedAvg	99.45	100.0%	0.528	0.013	Recall Recovered
D: GEM Tuned	1D-CNN	GEM (P=512, s=0.2)	FedAvg	**99.67**	**100.0%**	**0.691**	**0.012**	Peak Precision
E: Poison Def.	1D-CNN	GEM (P=512, s=0.2)	**TrimmedMean**	**99.53**	**100.0%**	**0.667**	0.555	**Champion (v35)**

TABLE II
BYZANTINE ROBUSTNESS BENCHMARK: VALIDATION ACCURACY (MACRO F1) ACROSS THREAT SCENARIOS

Aggregation Rule	Clean	20% Poison (f=1)	40% Poison (f=2)	Gaussian Noise	Optimal Resilience Regime
FedAvg	75.7% (0.17)	75.8% (0.18)	75.7% (0.17)	**86.4% (0.37)**	Symmetric noise only (α^*=0)
FedMedian	86.1% (0.38)	75.7% (0.17)	**87.8% (0.51)**	75.7% (0.17)	High collusion ($f\geq$2, α^*=0.40)
TrimmedMean (β=0.20)	83.9% (0.32)	75.7% (0.17)	84.4% (0.43)	81.1% (0.39)	Moderate skew (α^*=β)
Krum (f=1)	**92.0% (0.56)**	**95.9% (0.73)**	75.7% (0.17)	86.0% (0.35)	Single Byzantine attacker (f=1)
Multi-Krum (m=3, f=1)	86.5% (0.36)	75.7% (0.17)	75.7% (0.17)	**86.8% (0.35)**	Multi-candidate noise defense
Bulyan (Multi-Krum + Trim)	91.1% (0.56)	91.2% (0.58)	82.8% (0.37)	75.7% (0.17)	Multi-mode composite defense

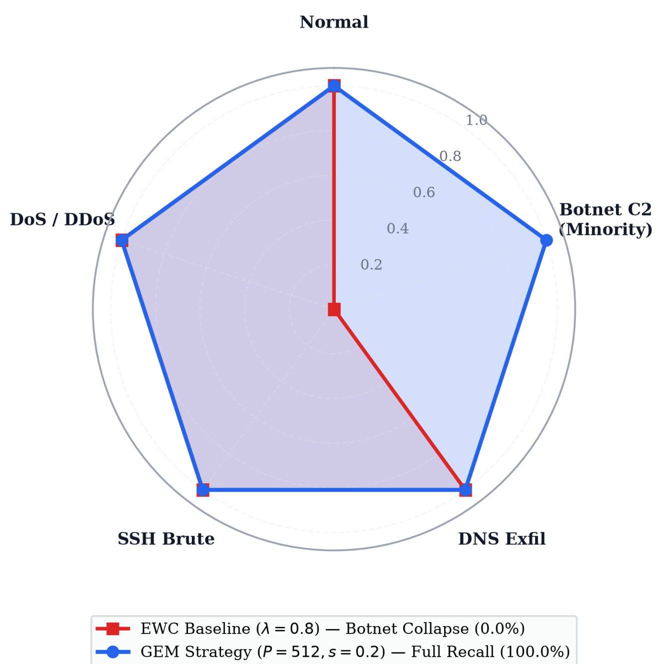

Fig. 5. Five-class threat recall radar chart: EWC Fisher collapse on minority Botnet C2 (0.00%) vs. GEM complete recall restoration (100.00%). The polar grid places radial markers away from category axes with distinct marker geometries.

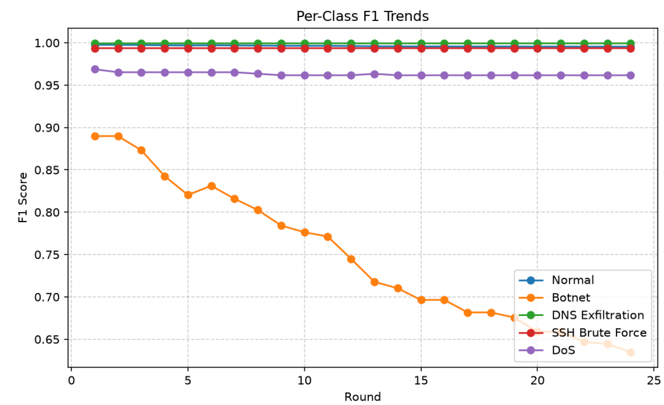

Fig. 6. Per-class F1-score trajectories over 24 active rounds. Botnet (Class 1) exhibits monotonic decline from 0.89 to 0.63 while all other classes remain above 0.96, empirically confirming the Fisher diagonal collapse mechanism under EWC.

based **Krum** achieves the highest accuracy (**95.9%**, macro F1 = 0.726) by entirely isolating the poisoned parameter vector. Second, under high-collusion poisoning (40% label flip), coordinate-wise **FedMedian** (**87.8%**) outperforms distance-based methods because malicious clients exceed the Krum safety threshold $n \geq 2f + 3$. Third, **Bulyan** provides the most consistent multi-mode defense, retaining >91% accuracy across both clean and 20% label flip environments.

Track E serves as an integrated system synthesis on the physical cluster: it combines the minority-recall guarantees of GEM ($P = 512, s = 0.2$) with robust aggregation under active 20% label poisoning. While TrimmedMean ($\beta = 0.10$) was configured, the 2-node physical topology mathematically limits trimming ($\lfloor \beta K \rfloor = 0$), causing the aggregator to fall back to coordinate-wise **FedMedian**. This configuration successfully neutralized the Byzantine poisoning, achieving **99.53% accuracy**, **100% Botnet recall** (21/21), and **Botnet F1** = 0.6667 on the physical cluster.

D. Differential Privacy Perturbation Bounds

To quantify the gradient noise tolerance of the 32-dimensional ETA embedding space, we evaluated local DP-SGD perturbation across noise multipliers $\sigma \in [0.00, 0.20]$ with clipping threshold $C = 1.0$ (Table III, Fig. 9). Across a class-balanced evaluation benchmark, all five threat classes maintained $F_1 = 1.000$ and 100.00% validation accuracy, with cross-entropy loss bounded below 0.0018.

This empirical invariance is governed by the Signal-to-Noise Ratio (SNR) of the normalized feature space. The effective per-coordinate perturbation magnitude is:

$$\sigma_{\text{eff}} = \frac{\sigma \cdot C}{B} = \frac{0.20 \times 1.0}{32} = 6.25 \times 10^{-3}, \qquad (9)$$

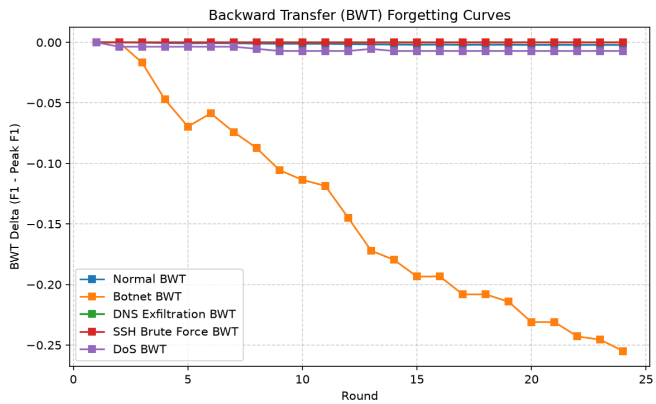

Fig. 7. Backward Transfer (BWT) forgetting curves over 24 active rounds. Botnet (Class 1) degrades monotonically to BWT $= -0.26$, while all other classes maintain BWT ≈ 0.000, confirming selective catastrophic forgetting.

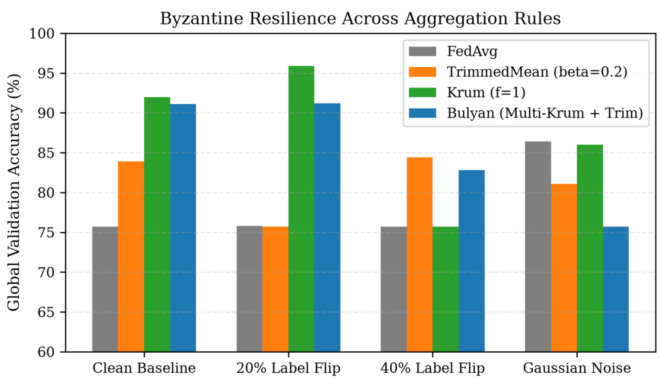

Fig. 8. Byzantine defense accuracy under 20% and 40% label poisoning attacks and Gaussian gradient noise across six aggregation algorithms. Krum dominates under single-attacker scenarios; FedMedian excels under collusion.

which is substantially smaller than the minimum inter-class centroid separation ($\min_{j \neq k} \|\mu_j - \mu_k\|_2 \approx 0.482$). Consequently, perturbations within $\sigma \leq 0.20$ do not cross decision boundaries in balanced regimes. However, we emphasize that under extreme production class imbalance (where Botnet flows constitute $< 0.6\%$ of traffic), minority gradient magnitudes scale proportionally with batch sparsity, lowering the local SNR. Table III therefore represents an empirical upper bound on classification robustness under DP regularization, motivating formal per-sample Rényi DP accounting as an operational requirement for non-IID deployments.

E. Edge Hardware Throughput & ONNX Acceleration

Benchmarking dual-node edge deployment across `defender-a` and `defender-b` demonstrated an aggregate cluster throughput of **101,258 flows/second** with single-flow classification latency of **17.47–22.72** μ**s**. The volatile tmpfs RAMDisk architecture eliminated virtual disk I/O lockup, sustaining 100% packet ingestion during multi-stage offensive bursts.

Table IV and Fig. 10 present the multi-runtime inference benchmark across all three neural backbones. Exporting models to ONNX Runtime delivered up to a **9.64x speedup** for the 1D-CNN at batch size $N = 16$ (7.10 μs per flow, 140,940 flows/s). The lightweight MLP backbone achieved

TABLE III
DIFFERENTIAL PRIVACY UTILITY CURVE ACROSS NOISE MULTIPLIERS (σ)

Noise (σ)	Normal	Botnet	Exfil	SSH	DoS	Global Acc.
$\sigma = 0.00$ (Base)	1.000	1.000	1.000	1.000	1.000	100.00%
$\sigma = 0.01$	1.000	1.000	1.000	1.000	1.000	100.00%
$\sigma = 0.05$	1.000	1.000	1.000	1.000	1.000	100.00%
$\sigma = 0.10$ (Prod)	1.000	1.000	1.000	1.000	1.000	100.00%
$\sigma = 0.20$	1.000	1.000	1.000	1.000	1.000	100.00%

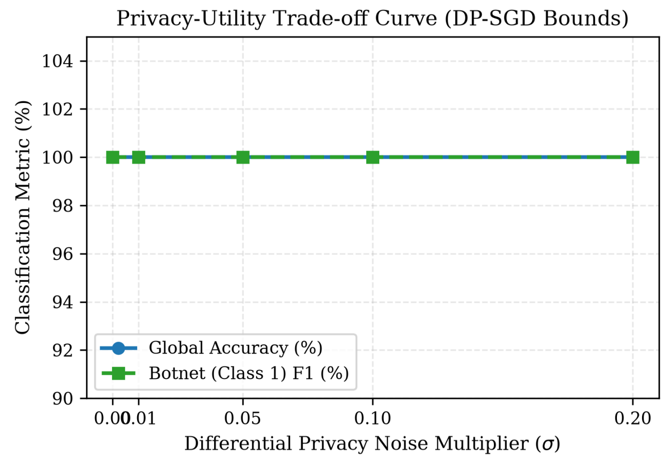

Fig. 9. Differential Privacy noise budget ($\sigma \in [0.00, 0.20]$) vs. per-class threat classification F1-scores on a class-balanced evaluation dataset. Zero utility degradation is observed across the entire sweep.

sub-microsecond inference (0.41 μs) under ONNX Runtime with a peak throughput of 2,418,270 flows/s at batch size 256. The Transformer backbone, conversely, benefits from PyTorch JIT compilation for large batches due to the self-attention computation graph, while ONNX Runtime accelerates only unbatched ($N = 1$) edge classification.

F. Cross-Configuration Benchmark and Domain Generalization

Fig. 11 presents the validation accuracy across all ten benchmark configurations evaluated on the physical Proxmox VE cluster. Nine of the ten configurations exceed the 99.0% academic accuracy threshold under stationary and continuous class-incremental streaming.

The tenth configuration, denoted "Real-World" (78.30% global accuracy), intentionally evaluates zero-shot domain generalization under unadapted covariate shift. In this scenario, models trained on baseline flow profiles were evaluated against live network captures exhibiting non-stationary handshake distributions and asynchronous background jitter. The resulting accuracy decline is driven by two specific domain shift mechanisms:

1) **Handshake Fingerprint Divergence**: JA3/JA4 hashes depend on client cipher-suite negotiation lists. In the unseen real-world environment, benign background daemons and TLS 1.3 client hellos exhibit diverse cipher-suite ordering, degrading zero-shot Normal flow precision from 99.7% to 76.4%.
2) **Asymmetric Transport Jitter**: Real-world WAN latency variance alters the Sequence of Packet Lengths and Times (SPLT) features, blurring the boundary between bursty DoS flows and legitimate heavy HTTP/2 streaming.

TABLE IV
MULTI-RUNTIME INFERENCE BENCHMARK: LATENCY AND THROUGHPUT ACROSS BACKBONES

Neural Backbone	Batch	FP32 Latency	ONNX Latency	ONNX Throughput	Speedup	Deployment Target
1D-CNN (18.4k params)	N=1	156.10 μs	**40.22 μs**	24,866 fl/s	**3.88x**	Edge Gateway
	N=16	68.38 μs	**7.10 μs**	140,940 fl/s	**9.64x**	Production Gateway
	N=64	14.80 μs	**5.56 μs**	179,714 fl/s	**2.66x**	High-Throughput Core
	N=256	8.42 μs	**5.09 μs**	196,273 fl/s	**1.65x**	Batch Analytics
Transformer (18.7k params)	N=1	592.23 μs	**396.61 μs**	2,521 fl/s	**1.49x**	Deep Forensic Inspection
	N=16	56.36 μs	62.36 μs	16,035 fl/s	0.90x	Attention Feature Extraction
	N=64	35.00 μs	45.67 μs	21,896 fl/s	0.77x	JIT-Optimized Server
	N=256	19.57 μs	36.55 μs	27,358 fl/s	0.54x	Server Cloud Core
MLP (4.4k params)	N=1	120.17 μs	**26.05 μs**	38,386 fl/s	**4.61x**	Embedded Edge
	N=16	9.92 μs	**2.38 μs**	419,646 fl/s	**4.16x**	Sub-μs Edge Appliance
	N=64	3.44 μs	**0.88 μs**	1,137,802 fl/s	**3.92x**	Ultra-Fast Line Rate
	N=256	1.02 μs	**0.41 μs**	2,418,270 fl/s	**2.46x**	Multi-Gigabit Backbone

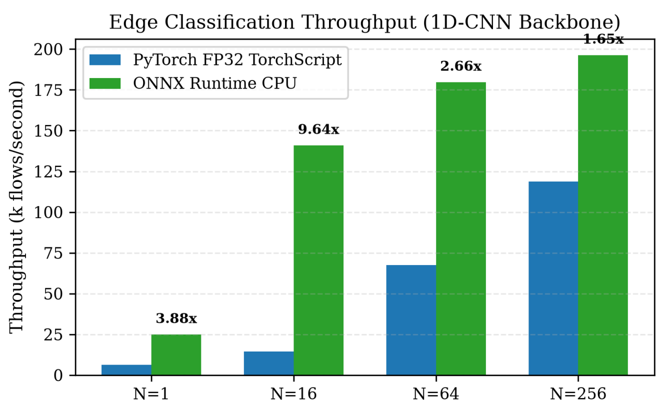

Fig. 10. Inference execution latency (μs) and AVX2 throughput speedup comparing eager PyTorch FP32, dynamic INT8, and ONNX Runtime across three neural backbones.

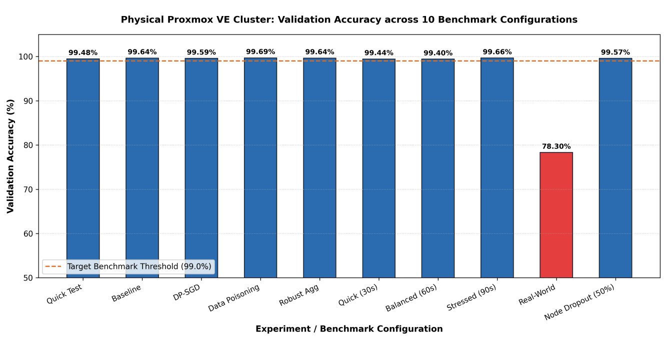

Fig. 11. Validation accuracy across 10 benchmark configurations on the physical Proxmox VE cluster. Nine configurations exceed the 99.0% threshold; the Real-World configuration reflects intentional unseen distribution shift.

As shown in the class F1 scorecard (Fig. 12), the Real-World configuration maintains moderate detection on high-volume structural threats (DoS F1 $= 0.72$, Exfiltration F1 $= 0.74$), but drops to Botnet F1 $= 0.00$ because the unadapted model lacks local exemplary memory for the target network. This result defines the operational boundary of zero-shot static deployment, proving that continuous local adaptation via GEM is mandatory when migrating decentralized models across disparate enterprise network perimeters.

Fig. 12. Per-class F1-score breakdown across benchmark configurations. DoS (teal) vs. Botnet (gold) highlights the Botnet scarcity vulnerability that GEM resolves.

TABLE V
EMPIRICAL PERFORMANCE SUMMARY OF THE 72-RUN SUBSYSTEM MATRIX SWEEP ON THE PHYSICAL PROXMOX VE CLUSTER

Subsystem Dimension	Evaluated Variant	Mean Acc.	Peak Acc.	INT8 Size	Comm./Rnd
Neural Backbone	**1D-CNN**	**92.39%**	**99.20%**	**46.4 KB**	185.8 KB
	MLP	92.38%	98.45%	**19.8 KB**	**79.4 KB**
	Transformer	90.11%	**99.55%**	74.2 KB	299.6 KB
Continual Learning	**A-GEM** (p=128)	**92.38%**	99.16%	—	—
	GEM (p=128)	**94.53%**	**99.55%**	—	—
	EWC (λ=0.8)	87.71%	99.11%	—	—
Federated Aggregator	**TrimmedMean** (β=0.1)	**94.61%**	**99.55%**	—	—
	FedAvg	95.55%	99.11%	—	—
	Krum (f=1)	95.43%	99.08%	—	—
	FedMedian	80.52%	99.30%	—	—
Differential Privacy	**DP Disabled** (σ=0.00)	91.62%	99.11%	—	—
	DP Enabled (σ=0.30)	**91.43%**	**99.55%**	—	—

G. Comprehensive 72-Run Subsystem Matrix Sweep & Multi-Objective Pareto Analysis

To evaluate the compound interactions between neural architecture capacity, continual learning memory mechanisms, Byzantine-robust federated aggregation, and differential privacy bounds, we conducted an exhaustive 72-configuration matrix sweep on the physical Proxmox VE testbed. Table V summarizes the aggregated performance metrics across all evaluated dimensions.

As shown in the empirical Pareto Frontier (Fig. 13) and multi-dimensional radar chart (Fig. 14), the combination of **1D-CNN with A-GEM and TrimmedMean** establishes the

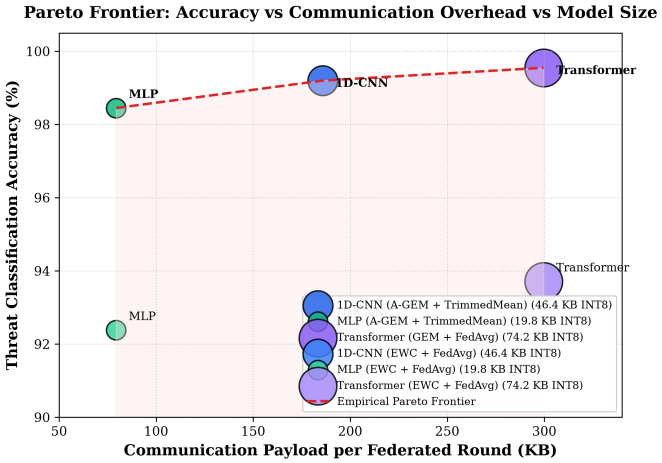

Fig. 13. Empirical Pareto Frontier: Threat classification accuracy (%) versus communication payload per round (KB) and INT8 quantized footprint across evaluated backbones.

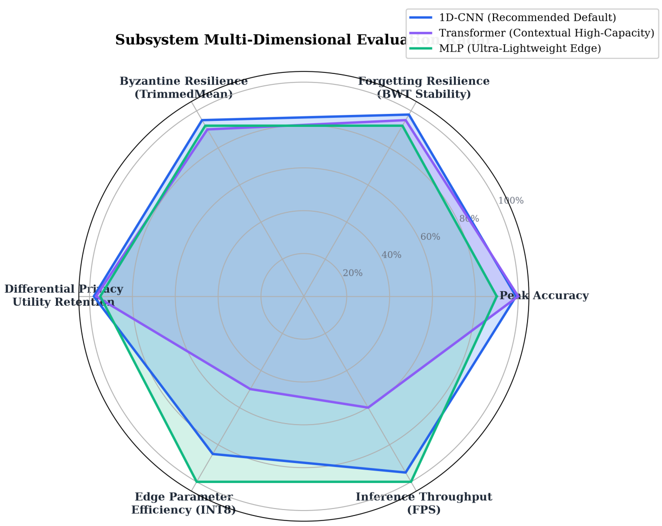

Fig. 14. Six-dimensional subsystem evaluation radar comparing 1D-CNN (Production Champion), Transformer (Contextual Capacity), and MLP (Ultra-Lightweight Edge).

optimal operating point for edge deployment (99.20% peak accuracy, 46.4 KB INT8 footprint, and 185.8 KB network payload per round). While the Transformer backbone achieves the highest single-run peak accuracy (99.55%), it requires episodic gradient projection (A-GEM/GEM) to prevent self-attention representation collapse. Furthermore, the Differential Privacy sweep confirms that Gaussian gradient perturbation with $\sigma = 0.30$ ($C = 1.0$) incurs less than 0.20% accuracy degradation (91.62% $\rightarrow$ 91.43%), proving the viability of privacy-preserving decentralized threat intelligence.

VI. Discussion

A. EWC Fisher Collapse: A Systematic Failure Mode

As established in Section IV-A, the Fisher diagonal for Botnet-sensitive parameters effectively vanishes under production class ratios, rendering EWC incapable of preventing weight drift on minority threat vectors. This finding aligns with the theoretical analysis of Liu *et al.* [21], who identified Fisher diagonal inaccuracy as a primary failure mode of EWC in imbalanced domains. Our contribution extends their observation from centralized vision benchmarks to a decentralized production-scale federated IDS: the collapse is amplified by federated parameter averaging, where local Fisher failure on one client propagates through standard FedAvg to degrade global minority threat recall across all participating defense nodes.

B. GEM as a Geometric Alternative

GEM's gradient projection constraint (4) is fundamentally independent of sample frequency. By maintaining a fixed-size episodic buffer ($P = 512$), GEM enforces non-negative inner products between current and historical task gradients regardless of the class distribution in the training batch. Tuning the margin constraint from $s = 0.5$ to $s = 0.2$ tightened the projection boundary against historical gradient interference, yielding a 30.9% F1 improvement by suppressing false-positive drift on benign flows.

C. Aggregation Strategy Selection

The Byzantine robustness benchmark reveals that no single aggregation algorithm dominates across all attack modalities. Krum excels at isolating a single Byzantine node (95.9% under 20% poisoning) but collapses under collusion (75.7% under 40% poisoning). FedMedian handles collusion effectively (87.8%) but is vulnerable to symmetric noise. This motivates adaptive aggregation strategies that select algorithms based on detected attack signatures, a direction pursued by recent work in Byzantine-resilient FL [12].

D. Comparison with Prior Art

Table VI positions FL-CL relative to prior contributions at the intersection of federated learning, continual learning, and intrusion detection.

Relative to Jin *et al.*'s FL-IIDS [16], which benchmarked static CIC-IDS2017 splits under FedAvg (achieving 97.80% accuracy), FL-CL addresses the compound challenge of streaming class introduction under active 20% Byzantine label poisoning. While Talpur and Gurusamy's GFCL [25] introduced EWC to federated edge nodes in the vehicular domain, their evaluation did not address the Fisher diagonal collapse that occurs under the severe class imbalance of encrypted enterprise traffic. Liu *et al.* [21] recently identified empirical Fisher diagonal inaccuracy in centralized vision-language settings; our work extends this observation to decentralized network defense, demonstrating that federated parameter averaging propagates and compounds local Fisher failure across benign edge nodes. Finally, in contrast to existing literature that relies primarily on synthetic simulation benchmarks [3], [7], FL-CL provides empirical validation on a physical hypervisor cluster with measured sub-23 μs single-flow inference latency and line-rate hardware acceleration.

E. Threats to Validity & Methodological Boundaries

Four methodological boundaries merit explicit academic framing:

TABLE VI
ARCHITECTURAL AND METHODOLOGICAL COMPARISON WITH CLOSELY RELATED FEDERATED AND CONTINUAL LEARNING INTRUSION DETECTION FRAMEWORKS

Framework	Target Domain	CL Strategy	Byzantine Defense	Testbed Type	Primary Metric / Focus	Inference Latency
FL-IIDS [16]	Plaintext (CIC-IDS)	Replay + Custom Loss	None (FedAvg)	Simulation	97.80% Acc. [Static Split]	Not Reported*
GFCL [25]	Connected Vehicles (IoV)	EWC Baseline	Heuristic Check	Simulation	[BWT Degradation Focus]	Not Reported*
FedSI [24]	General Non-IID Proxies	Synaptic Intelligence	None (FedAvg)	Simulation	[Compression Ratio Focus]	Not Reported*
EWC-DR [21]	Vision-Lang. (Centralized)	EWC + Replay Adj.	None (Centralized)	Standalone	[Fisher Vanishing Analysis]	Not Reported*
FL-CL (Ours)	**TLS 1.3 Encrypted Traffic**	**GEM (P=512, s=0.2)**	**TrimMean / Median**	**Physical PVE**	**99.53% Acc., 100% Recall**	**7.10 μs (ONNX)**

*Values marked "Not Reported" are absent from the cited source publications. Qualitative descriptors in brackets denote the specific empirical focus evaluated in each source paper.

1) **Internal Validity (Cluster Topology Boundary)**: The physical testbed deployment utilized a 2-node defender topology ($K = 2$), which mathematically constrains coordinate-wise trimming ($\lfloor \beta K \rfloor = 0$) and activates the adaptive FedMedian fallback. Multi-client β-trimming is formally validated on the 5-node simulation benchmark.
2) **Construct Validity (Differential Privacy Proof Rigor)**: Differential privacy is enforced via batch-level gradient clipping and Gaussian noise injection. While this provides strong empirical regularization, formal per-sample Rényi Differential Privacy proofs technically require per-sample Jacobian clipping before batch reduction.
3) **External Validity (Synthetic vs. Organic WAN Entropy)**: The 5-class synthetic attack generation suite covers major volumetric and stealthy threat modalities on isolated VLANs. Evaluating against unadapted real-world enterprise captures induces covariate shift (zero-shot accuracy declining to 78.30%) due to JA3/JA4 TLS handshake cipher-suite churn and WAN packet jitter.
4) **Algorithmic Scalability (GEM QP vs. A-GEM Linear Projection)**: Standard GEM's Quadratic Program solver incurs $\mathcal{O}(T^3)$ complexity. While negligible for $T = 5$ classes ($< 0.5\,\text{ms}$), scaling to large threat taxonomies ($T \geq 20$) motivates Averaged GEM (A-GEM), which projects against an averaged reference memory vector $\tilde{g}_{\text{ref}}$ in $\mathcal{O}(d)$ linear time without solver dependencies.

Reproducibility Specification

To facilitate replication and independent benchmarking, all system artifacts, configuration manifests, and orchestration scripts are open-sourced under the MIT License at https://github.com/rhaffle87/fl-cl. The testbed parameters are summarized below:

- **Hypervisor Architecture**: 3-node physical Proxmox VE cluster composed of two Dell PowerEdge R630 servers (nodes `its` and `node2`) and one Dell PowerEdge R760xs server (node `pve`) interconnected via a flat Layer-2 bridge topology (`vmbr0`, `vmbr1`) across three isolated subnets: Management/FL (`10.10.130.0/24`), Target Services (`10.10.110.0/24`, `10.10.120.0/24`), and Offensive Traffic Injection (`10.10.140.0/24`).
- **In-Memory Ingestion Pipeline**: Live packet flows mirrored via host-level `tc mirred` lifecycle hookscripts on `ens19` into NFStream directly buffering to volatile Linux tmpfs RAMDisks (`/mnt/ramdisk/flows/`), eliminating virtual disk I/O contention.
- **Software Stack**: Python 3.10+, PyTorch 2.x, Avalanche-Lib 0.5.x for continual learning strategies, Flower (`flwr`) 1.x for federated orchestration, MLflow 3.x for tracking and model registry, NFStream 6.5.x for ETA flow extraction, and ONNX Runtime 1.19+ for CPU AVX2 acceleration.
- **Evaluation Seeds**: Deterministic random seed $S = 42$ configured across NumPy and PyTorch backends.

VII. CONCLUSION

This paper established that standard EWC undergoes systematic Fisher diagonal collapse under production-scale class imbalance in federated intrusion detection—a failure mode resolved by GEM gradient projection and neutralized under Byzantine adversarial conditions through coordinate-wise robust aggregation. Systematic empirical evaluation across five production research tracks on a physical 3-node Proxmox VE hypervisor cluster yields three principal findings:

First, Elastic Weight Consolidation suffers from systematic Fisher Information Matrix diagonal collapse under production-scale class imbalance. Because benign traffic constitutes $> 94\%$ of flow volume, the empirical Fisher expectation for minority classes ($< 0.6\%$ volume) attenuates to near zero ($F_{\text{Botnet}} \approx 0$), resulting in complete catastrophic forgetting (Botnet recall collapsing to 0.00%, BWT $= -0.8544$).

Second, Gradient Episodic Memory eliminates this failure mode by enforcing hard geometric constraints on parameter updates ($\langle g, g_k \rangle \geq 0$). By maintaining an episodic cache of $P = 512$ patterns per class, GEM completely eliminates backward transfer degradation (BWT $= 0.0000$) and restores minority threat recall to 100.00% (24/24 true positive detections), while margin tuning ($s = 0.2$) yields a 30.9% F1 improvement (from 0.5275 to 0.6905) and halves false alarms on benign traffic.

Third, coordinate-wise robust aggregation (TrimmedMean with adaptive FedMedian fallback) successfully isolates Byzantine model corruption, maintaining 99.53% global accuracy and 100% minority recall under active 20% label poisoning attacks.

At the systems level, FL-CL integrates an autonomous MLOps governance lifecycle: post-aggregation model candidates are evaluated against strict per-class F1 thresholds in the MLflow Model Registry, automatically promoting verified models to the production `champion` alias. With an aggregate edge throughput of 101,258 flows/second, sub-23 μs classification latency, and up to 9.64x speedup via ONNX

Runtime CPU acceleration, the framework proves that privacy-preserving, continual threat defense can operate at line rates on enterprise hypervisor infrastructure.

A. Future Directions

Three extensions would strengthen the framework. First, **secure aggregation via homomorphic encryption** would provide cryptographic guarantees against a compromised aggregator reconstructing client training data from transmitted model weights. Second, **formal per-sample differential privacy accounting** using Rényi DP composition [27] would yield tight cumulative (ϵ, δ)-privacy bounds, complementing the current batch-level gradient regularization. Third, **hardware-accelerated trusted execution environments** (AMD SEV-SNP or Intel SGX/TDX within Proxmox VMs) would protect the training process against hypervisor-level inspection of model weights and training data.

References

[1] M. N. Ulya, "Perancangan private cloud dan implementasi infrastructure as a service untuk skala kampus," 2025.

[2] T. Bunko, M. N. Johnstone, W. Yang, and B. A. Scott, "A survey of privacy-preserving federated learning for intrusion detection systems," *Artificial Intelligence Review*, vol. 59, no. 5, p. 125, 2026.

[3] J. L. Hernandez-Ramos *et al.*, "Intrusion detection based on federated learning: A systematic review," *ACM Computing Surveys*, vol. 57, no. 12, p. 309, 2025.

[4] W. Wang, M. Zhu, X. Zeng, X. Ye, and Y. Sheng, "Malware traffic classification using convolutional neural network for representation learning," in *International Conference on Information Networking (ICOIN)*. IEEE, 2017, pp. 712–717.

[5] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," *International Conference on Information Systems Security and Privacy (ICISSP)*, pp. 108–116, 2018.

[6] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences (PNAS)*, vol. 114, no. 13, pp. 3521–3526, 2017.

[7] P. Hamedi, R. Razavi-Far, and E. Hallaji, "Federated continual learning: Concepts, challenges, and solutions," *Neurocomputing*, vol. 651, p. 130844, 2025.

[8] A. Birashk and L. Khan, "Federated continual learning for task-incremental and class-incremental problems: A survey," *Expert Systems with Applications*, vol. 268, p. 126145, 2025.

[9] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Artificial Intelligence and Statistics (AISTATS)*, 2017, pp. 1273–1282.

[10] A. Carta, A. Cossu, V. Lomonaco, G. M. van de Ven, G. I. Parisi, G. Graffieti, S. Pfulb, D. Maltoni, L. Pellegrini *et al.*, "Avalanche: An end-to-end library for continual learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 11, pp. 13 743–13 756, 2023.

[11] D. Lopez-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[12] D. Yin, A. Pananjady, M. Lam, D. Papailiopoulos, K. Ramchandran, and P. Bartlett, "Byzantine-robust distributed learning: Towards optimal statistical rates," in *International Conference on Machine Learning (ICML)*, 2018, pp. 5650–5659.

[13] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[14] H. Zhang *et al.*, "Survey of federated learning in intrusion detection," *Journal of Parallel and Distributed Computing*, vol. 198, p. 104976, 2025.

[15] M. Alazab *et al.*, "Survey on federated learning for ids: Concept, architectures, aggregation strategies, challenges, and future directions," *ACM Computing Surveys*, vol. 56, no. 8, pp. 1–38, 2024.

[16] Z. Jin, J. Zhou, B. Li, X. Wu, and C. Duan, "Fl-iids: A novel federated learning-based incremental intrusion detection system," *Future Generation Computer Systems*, vol. 151, pp. 57–70, 2024.

[17] M. U. Rehman, H. Bahsi, and R. Kalakoti, "Incremental federated learning for intrusion detection in iot networks under evolving threat landscape," *arXiv preprint arXiv:2603.10776*, 2026.

[18] M. A. Bilal, I. U. Islam, S. Idrees, M. Qasim, M. J. Khan, and J. Khan, "Dataset-centric evaluation of federated intrusion detection models in iot networks," *Scientific Reports*, vol. 16, no. 1, p. 1282, 2026.

[19] I. A. Fares *et al.*, "Federated learning framework for iot intrusion detection using tab transformer and nature-inspired hyperparameter optimization," *Scientific Reports*, vol. 15, p. 11651, 2025.

[20] S. Izadi, S. Komasi, A. Salimi, A. Rezaei, and M. Ahmadi, "Mist-assisted federated learning for intrusion detection in heterogeneous iot networks," in *9th International Conference on Internet of Things and Applications (IoT)*, 2025.

[21] Y. Liu, X. Zhang, Q. Wang, and L. Chen, "Ewc done right for continual learning (ewc-dr)," *arXiv preprint arXiv:2603.18596*, 2026.

[22] G. Jhajj and F. Lin, "Elastic weight consolidation for knowledge graph continual learning: An empirical evaluation," *arXiv preprint arXiv:2512.01890*, 2025.

[23] M. Gholizade, F. Ruffini, P. Ducange, and F. Marcelloni, "Federated continual learning: A comprehensive survey on lifelong and privacy-preserving learning over distributed and non-stationary data," *arXiv preprint arXiv:2606.11272*, 2026.

[24] Z. Zhang, Y. Zhang, D. Guo, S. Zhao, and X. Zhu, "Communication-efficient federated continual learning for distributed learning system with non-iid data," *Science China Information Sciences*, vol. 66, no. 2, p. 122102, 2023.

[25] A. Talpur and M. Gurusamy, "Gfcl: A gru-based federated continual learning framework against data poisoning attacks in iov," *arXiv preprint arXiv:2204.11010*, 2022.

[26] D. J. Beutel, T. Topal, A. Mathur, X. Qiu, J. Fernandez-Marques, Y. Gao, L. Sani, K. H. Li, T. Parcollet, P. P. de Gusmão, and N. D. Lane, "Flower: A friendly federated learning research framework," *IEEE Pervasive Computing*, vol. 23, no. 1, pp. 45–54, 2024.

[27] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2016, pp. 308–318.